

MATH 448-548

FALL 2025 SYLLABUS

Location and time: 10:35-11:50 T, Th. WLSH 13.

Instructor: Prof. Alexander Panchenko, panchenko@wsu.edu

Office: Neill 329.

Phone: 335-3127.

OFFICE HOURS: TBA

Textbook: *Numerical Analysis* by D. Kincaid and W. Cheney, AMS Publishing., 3d edition.

Coverage: We shall cover parts of ch. 1-9.

Note: *some cuts and additions are possible.*

Topics covered: Solution of nonlinear equations; Systems of linear equations; Approximating functions; Numerical integration and differentiation; Numerical solution of ordinary and partial differential equations;

Student Learning Outcomes: Mastery of basics of numerical analysis. Students should be able to understand the details of algorithm design, error analysis, and stability analysis. Familiarity with MATLAB and R programming will be also developed.

GRADING POLICY

There will be one midterm project and a final project. Both of these projects will be take-home. Your homework will be collected approximately once a week. Your grade will be determined by the midterm project (20 %), HW (40%) and the final (40 %). **No assignments that are more than one week past due will be accepted. Assignments should be submitted online into Canvas.**

Use of AI is prohibited. Students are not allowed to use advanced automated tools (artificial intelligence or machine learning tools such as ChatGPT, Co-Pilot, or Dall-E) on assignments in this course. Each student is expected to complete each assignment without substantive assistance from others, including automated tools.

Web Site: the website for the class is maintained in Canvas canvas.wsu.edu. Throughout the semester, all relevant materials will be posted and the announcements will be also distributed by email. Please make sure that a valid email address is on file at the Registrar's office.

STUDY ASSISTANCE AND TUTORING OPPORTUNITIES. Successful students make use of available resources, so don't struggle when help is just a few steps away. We want you to succeed, we're here for you, and our tutoring services are completely FREE.

The Math Learning Center (MLC) will continue to offer virtual tutoring via Zoom this semester. You can find current information on how to connect with a tutor on our website:
<http://www.math.wsu.edu/studyhalls/welcome.php>

In addition to virtual tutoring, the MLC will also host a forum where students can post questions to be addressed by our MLC tutors as well as provide an opportunity to connect with other students taking the same course. Visit <https://mathlab.math.wsu.edu/phpBB3> and create an account to start connecting with our tutors. (Note: This is a monitored forum. Please read the guidelines posted on the forum before posting.)

Come see me during office hours! I am here to help you. It is my goal to see you succeed in this class and future classes.

ENGLISH WRITING LAB. Writing proofs is more of a writing exercise than a math exercise. WSU does provide a writing lab to assist: <https://writingprogram.wsu.edu/undergraduate-writing-center/>. They may not be able to help with the math, but they can help with the clarity.

TECHNOLOGY ISSUES. Do not wait until the last minute to complete online homework or access your course materials online. Technology fails so give yourself time to troubleshoot the problem and still meet the deadlines or work ahead when you can. You should be prepared with a backup plan. For each class meeting and assignment, some things to consider include:

- What would I do if the wifi failed?
- Does my printer have enough ink and paper?
- Are there certain times of the day that your internet speeds up or slows down?
- Are your devices fully charged?
- Are there other devices you can use if the current one fails? For technical issues that are WSU related, try getting help from <https://its.wsu.edu/csd>. I am here to help you with your math questions, however I am not a technical support person.

Campus Safety. Washington State University is committed to enhancing the safety of the students, faculty, staff, and visitors. It is highly recommended that you review the Campus Safety Plan (<http://safetyplan.wsu.edu/>) and visit the Office of Emergency Management web site (<http://oem.wsu.edu/>) for a comprehensive listing of university policies, procedures, statistics, and information related to campus safety, emergency management, and the health and welfare of the campus community.

Students with Disabilities: Reasonable accommodations are available for students with a documented disability. If you have a disability and need accommodations to fully participate in this class, please either visit or call the Access Center (Washington Building 217; 509-335-3417) to schedule an appointment with an Access Advisor. All accommodations MUST be approved through the Access Center. For more information contact a Disability Specialist on your home campus: Pullman or WSU Online: 509-335-3417 <http://accesscenter.wsu.edu>, Access.Center@wsu.edu

Academic integrity. Academic integrity is the cornerstone of the university. Any student who attempts to gain an unfair advantage over other students by cheating, will fail the assignment and be reported to the Office Student Standards and Accountability. Cheating is defined in the Standards for Student Conduct WAC 504-26-010 (3). Students who violate WSU Academic Integrity Policy (identified in Washington Administrative Code (WAC) 504-26-010(3) and -404) will receive a zero on the assignment or exam, and will not have the option to withdraw from the course pending an appeal, and will be reported to the Office of Student Conduct.

- Cheating includes, but is not limited to, plagiarism and unauthorized collaboration as defined in the Standards of Conduct for Students, WAC 504-26-010(3). You need to read and understand all of the definitions of cheating: <http://app.leg.wa.gov/WAC/default.aspx?cite=504-26-010>. If you have any questions about what is and is not allowed in this course, you should ask course instructors before proceeding.
- No form of cheating will be tolerated in this class. If caught cheating on any assignment or test, you and any other students involved will be given a zero on the assignment or test. Depending on the severity of the incident, you may also fail the course with no option to withdraw (see above). You will be reported to the Office of Student Conduct.
- I reserve the right to request an oral examination of any student, over any item. There will be some time given to schedule the oral exam but the exact nature of the questionable item(s) will not be shared. The grade for the oral exam will replace the grade for the questionable item(s), whether it improves or lessens the existing score. Failure of the student to meet in a reasonable time will score a zero on the questionable item(s).
- All course documents are copyrighted. It is prohibited for any student to copy, distribute or put online any course materials, including but not limited to lecture notes, assignments, exams, solution keys, etc. You cannot post the home-work or any course materials on a website such as Chegg.com or similar sites.
- If you wish to appeal a faculty member's decision relating to academic integrity, please use the form available at communitystandards.wsu.edu. Make sure you submit your appeal within 21 calendar days of the faculty member's decision.

COVID-19 POLICY AND STATEMENT. (Academic Regulation 72): Students are responsible for ensuring that they attend all class meetings and complete all in-class and out-of-class work as assigned by the instructor. Students are also responsible for communicating with the instructor as soon as possible should they need to be absent. Students should email the instructor from their WSU email address to notify the instructor of a long-term absence.

Students are expected to abide by all current COVID-19 related university policies and public health directives, which could include wearing a cloth face covering, physically distancing, self-attestations, and sanitizing common use spaces. All current COVID-19 related university policies and public health directives are located at <https://wsu.edu/covid-19>. Students who do not comply with these directives may be required to leave the classroom; in egregious or repetitive cases, students may be referred to the Center for Community Standards for university disciplinary action.

POLICY PROHIBITING DISCRIMINATION AND HARASSMENT. (Executive Policy 15): This policy expresses WSU's commitment to maintaining an environment free from discrimination, including sexual harassment. This policy applies to all students, faculty, staff, or others having an association with the University. Discrimination, including discriminatory harassment, sexual harassment, and sexual misconduct (including stalking, intimate partner violence, and sexual violence) is prohibited at WSU (See WSU Policy Prohibiting Discrimination and Harassment (Executive Policy 15) and WSU Standards of Conduct for Students (WAC Code 504-26)). If you feel you have experienced or have witnessed discriminatory conduct, you can contact the WSU Office of Civil Rights Compliance and Investigation (CRCI) and/or the WSU Title IX Coordinator at 509-335-8288 to discuss resources, including confidential resources, and reporting options. (Visit

crci.wsu.edu for more information). Most WSU employees, including faculty, who have information regarding sexual harassment or sexual misconduct are required to report the information to CRCI or a designated Title IX Coordinator or Liaison. Visit <https://crci.wsu.edu/reporting-requirements/> for more information.

ACCOMMODATION FOR RELIGIOUS OBSERVANCES OR ACTIVITIES. Washington State University reasonably accommodates absences allowing for students to take holidays for reasons of faith or conscience or organized activities conducted under the auspices of a religious denomination, church, or religious organization. Reasonable accommodation requires the student to coordinate with the instructor on scheduling examinations or other activities necessary for course completion. Students requesting accommodation must provide written notification within the first two weeks of the beginning of the course and include specific dates for absences. Approved accommodations for absences will not adversely impact student grades. Absence from classes or examinations for religious reasons does not relieve students from responsibility for any part of the course work required during the period of absence. Students who feel they have been treated unfairly in terms of this accommodation may refer to Academic Regulation 104 - Academic Complaint Procedures. See also Rule 82.

SEVERE WEATHER POLICY. For severe weather alerts, see: <http://alert.wsu.edu/> and <https://oem.wsu.edu/emergency-procedures/severe-weather/>. In the event of severe weather affecting university operations, guidance will be issued through the alert system.

STUDENTS IN CRISIS: PULLMAN RESOURCES.

If you or someone you know is in immediate danger, DIAL 911 FIRST!

Student Care Network: <https://studentcare.wsu.edu/>

Cougar Transit: 978 267-7233

WSU Counseling and Psychological Services (CAPS): 509 335-2159

Suicide Prevention Hotline: 800 273-8255

Crisis Text Line: Text HOME to 741741

WSU Police: 509 335-8548

Pullman Police (Non-Emergency): 509 332-2521

WSU Office of Civil Rights Compliance and Investigation: 509 335-8288

Alternatives to Violence on the Palouse: 877 334-2887

Pullman 24-Hour Crisis Line: 509 334-1133